



histopathology

Introduction to pathology

3rd class

Lab 1

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FIBMS. Pathology
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learning objectives

1. **Identify and explain** the primary subdivisions of clinical pathology (e.g., histopathology, cytopathology, hematopathology, microbiology) and state the main goals of pathology, including etiology and pathogenesis.
2. **Analyze and sequence** the sequential steps involved in histological tissue processing—from fixation, dehydration, clearing, and infiltration to embedding, sectioning with a microtome, and H&E staining.
3. **Compare and evaluate** the diagnostic applications, advantages, and limitations of histopathological tissue examination versus cytopathological techniques (such as Fine Needle Aspiration and exfoliative cytology).

Introduction to pathology

Pathology of any diseases mean structural changes, functional disturbance that presented as clinical features.

Pathosuffering

Logos Study of diseases

The goal of pathology :

- identification the causes of diseases (etiology).
- mechanism of the diseases (pathogenesis).
- assist in disease prevention and therapy.

It involves the examination of tissues, organs, bodily fluids and autopsies in order to study and diagnose disease



Subdivisions of clinical pathology

1. Histopathology
2. Cytopathology
3. Hematopathology
4. Microbiology
5. Immunopathology
6. Chemical pathology
7. Medical genetics
8. Toxicology
9. Forensix pathology



Histopathology

- Histopathology is the most significant branch of pathology.
- Histopathology focus on:
 1. examining tissues with the naked eye (gross examination)
 2. examining tissues under a microscope for definitive diagnosis of disease (light microscopy)

Microscope Parts

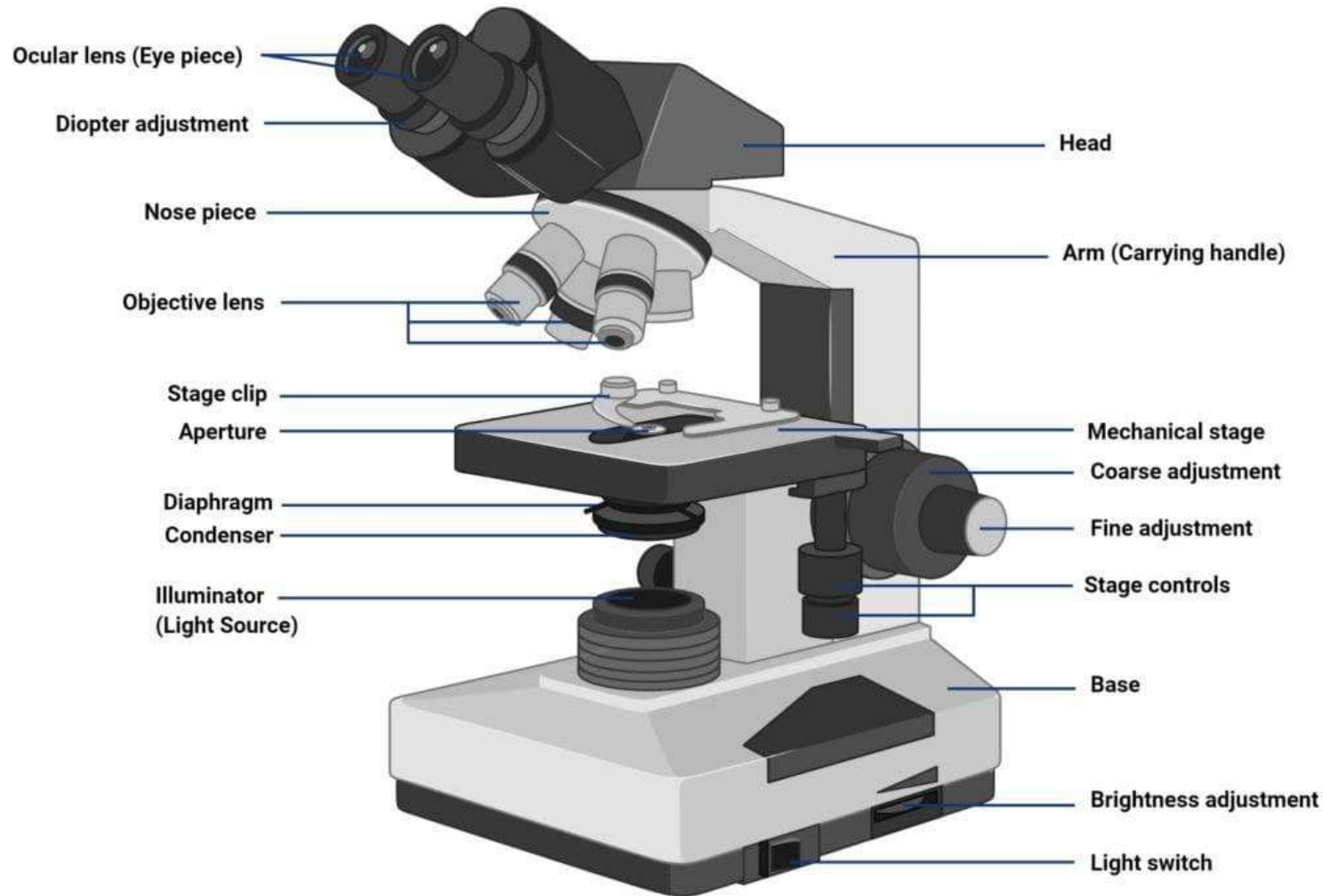
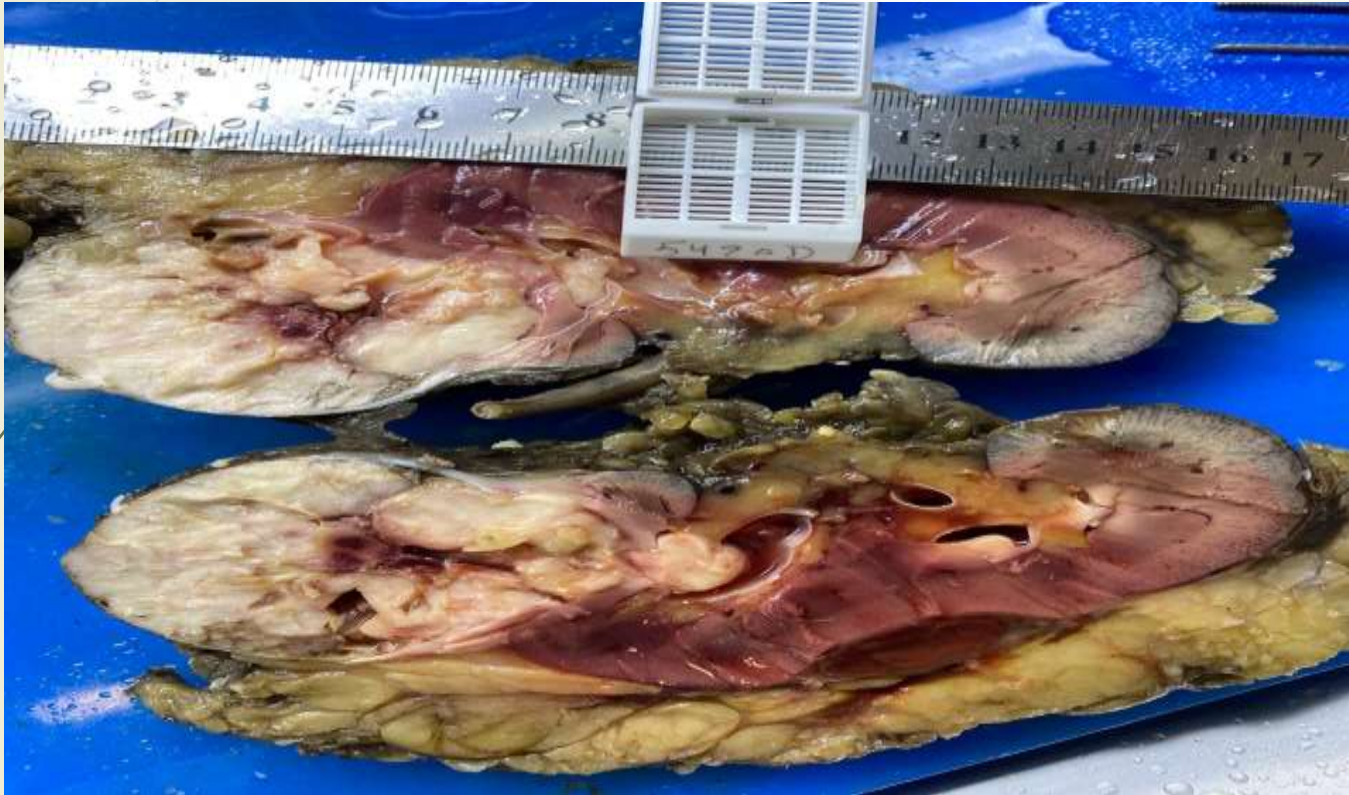


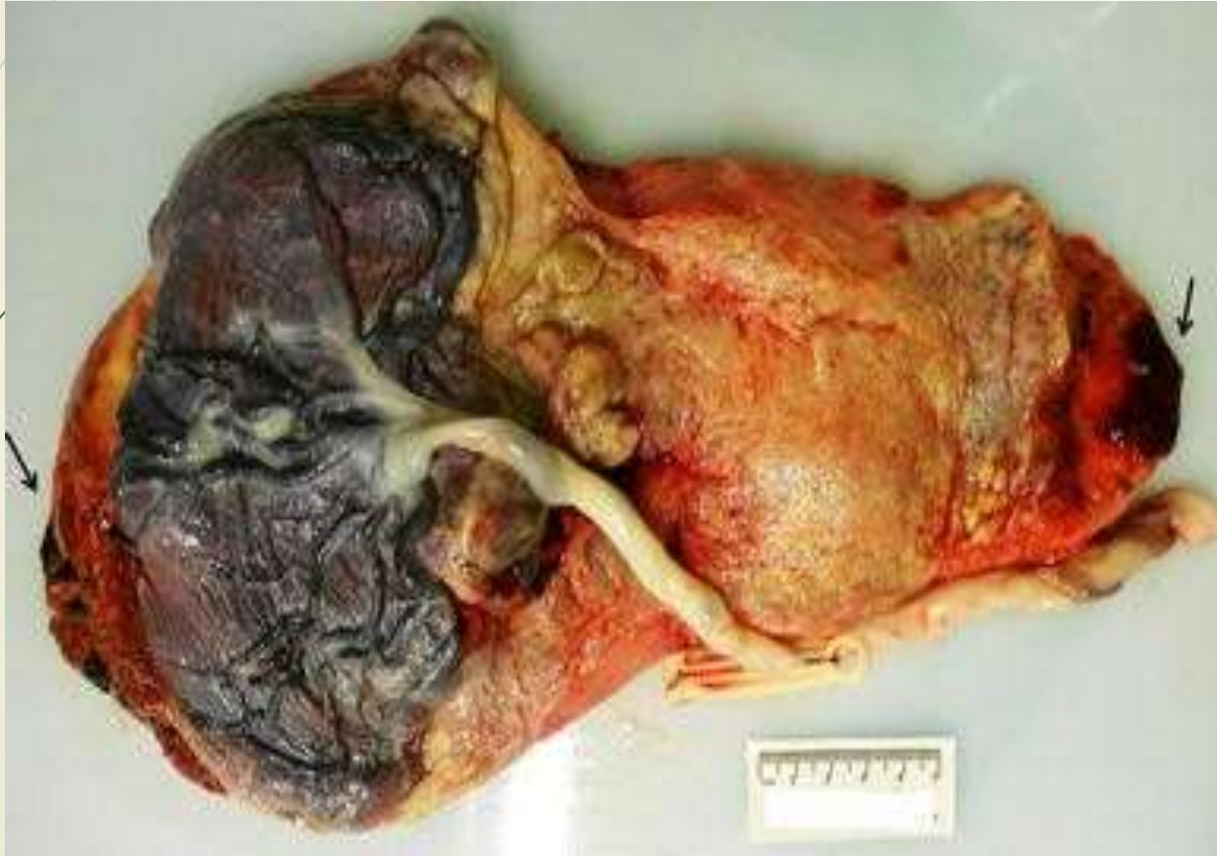
Figure: Parts of a microscope, Image Copyright © Sagar Aryal, www.microbenotes.com

kidney gross examination



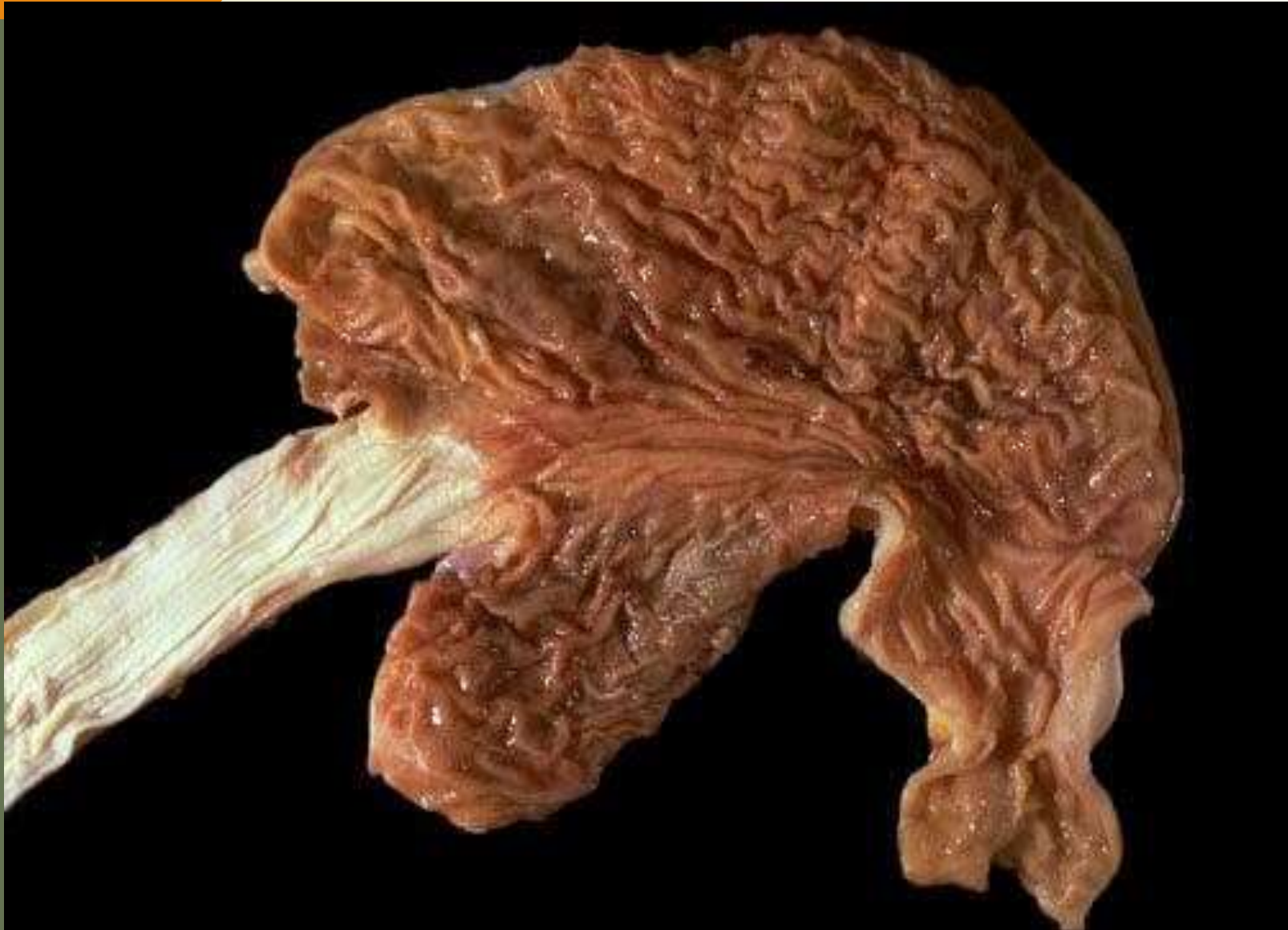
Abnormal white mass
present in the kidney

Uterus gross examination



Hysterectomy for
placenta accreta

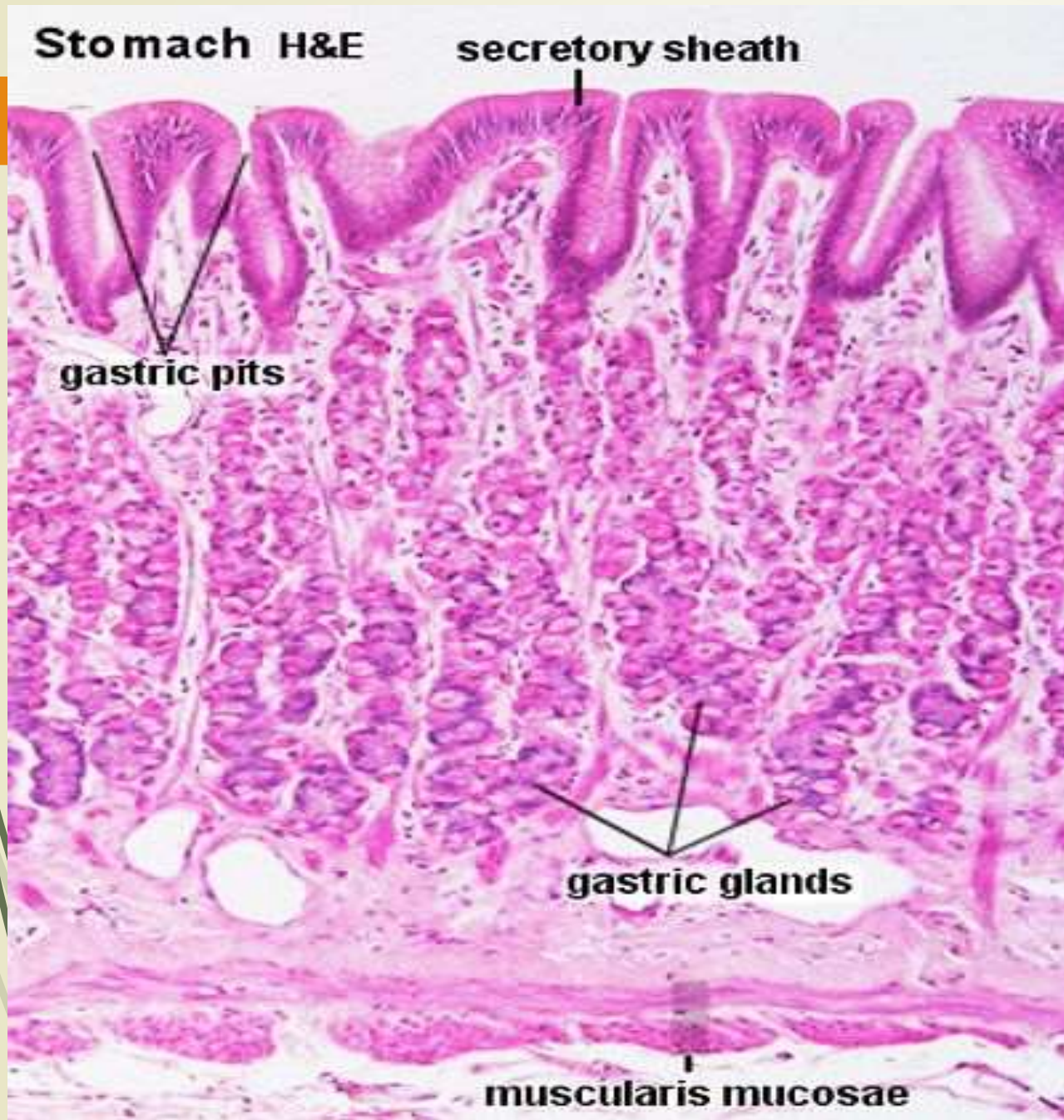
Stomach gross examination



This is the normal appearance of the stomach, which has been opened along the greater curvature.

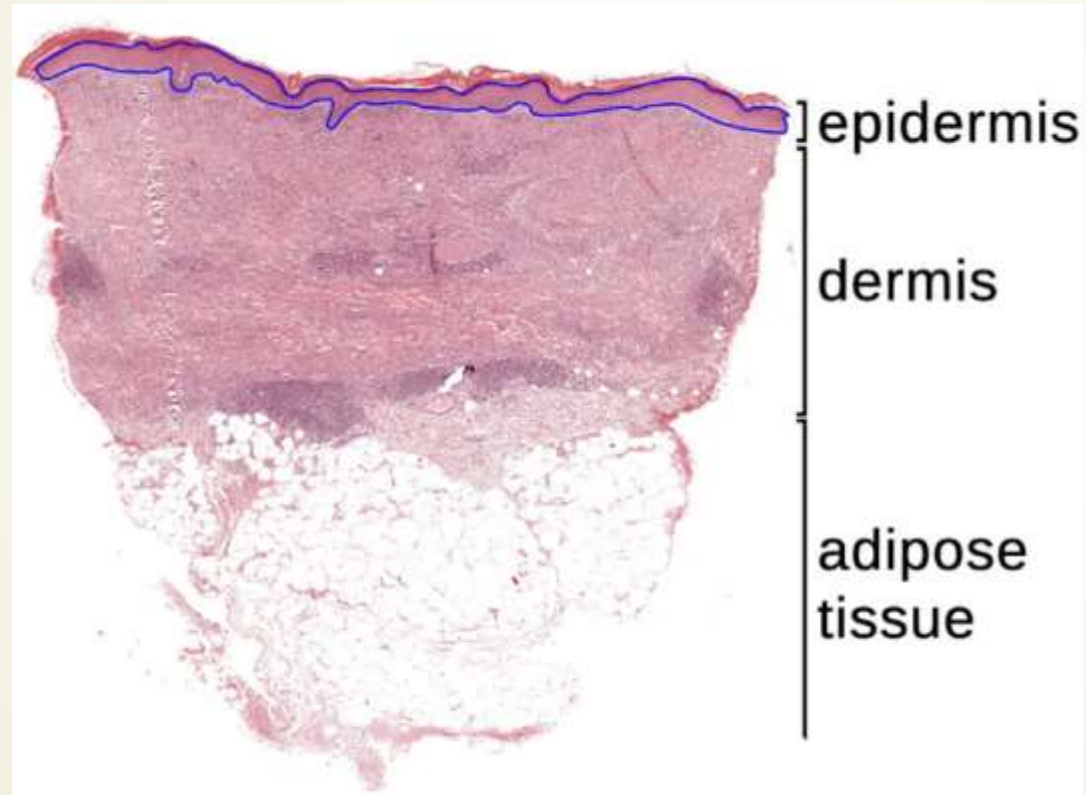
The esophagus is at the left.

Just beyond the antrum is the pylorus emptying into the first portion of duodenum is at the lower right.



Stomach under microscope

Microscopic picture of skin



Grossing hood

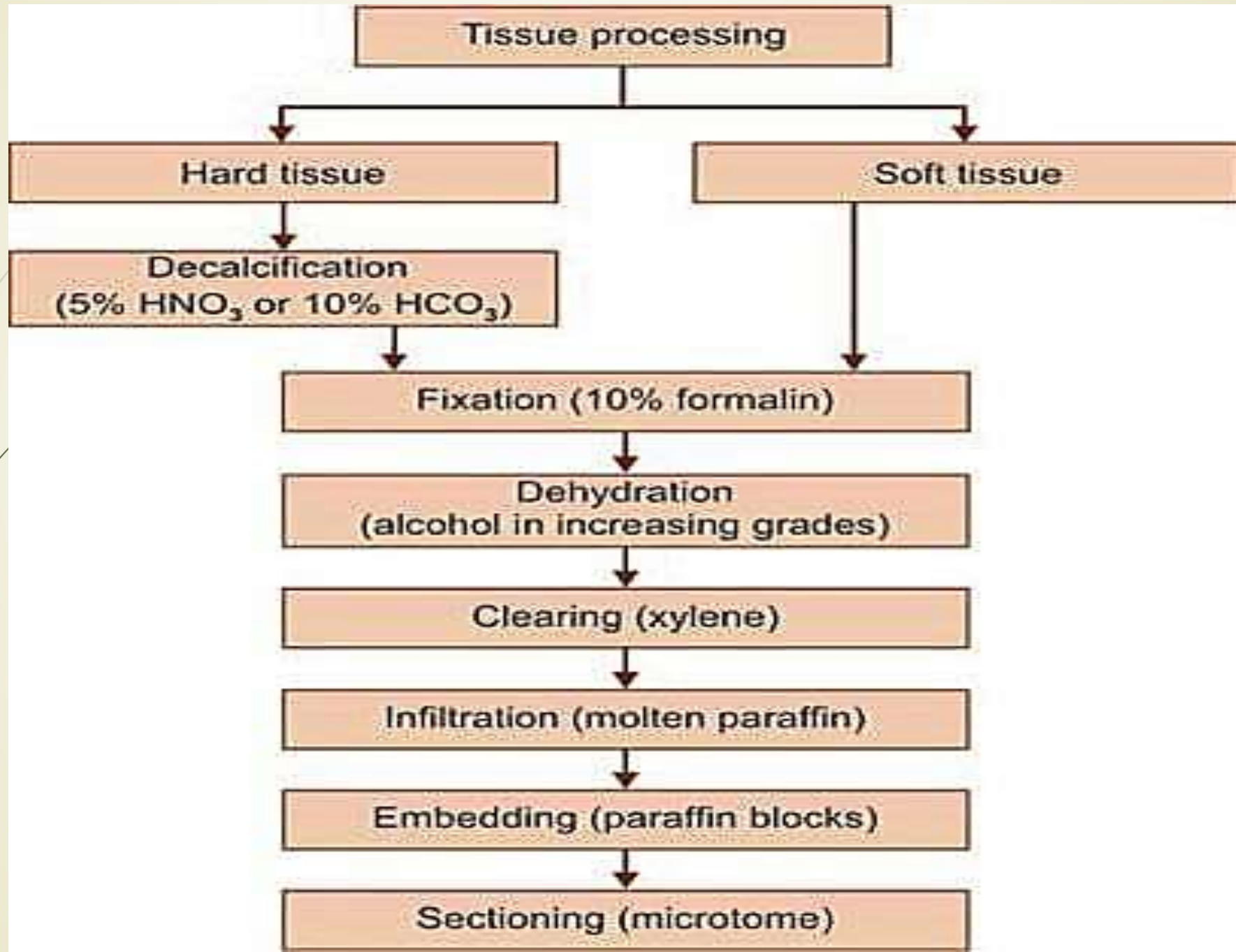


Tissue cassettes



Automated tissue processor

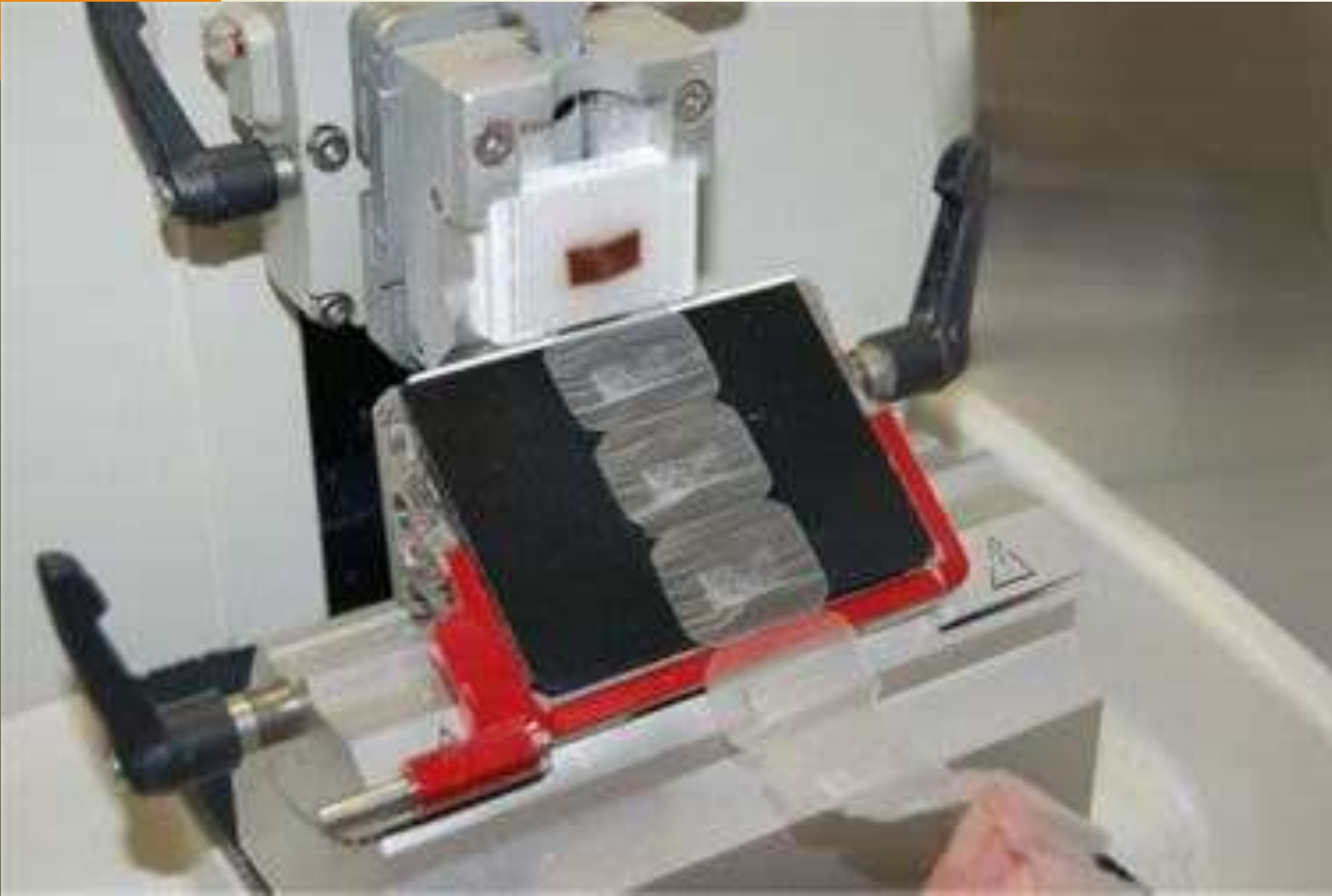






Paraffin embedded tissue block



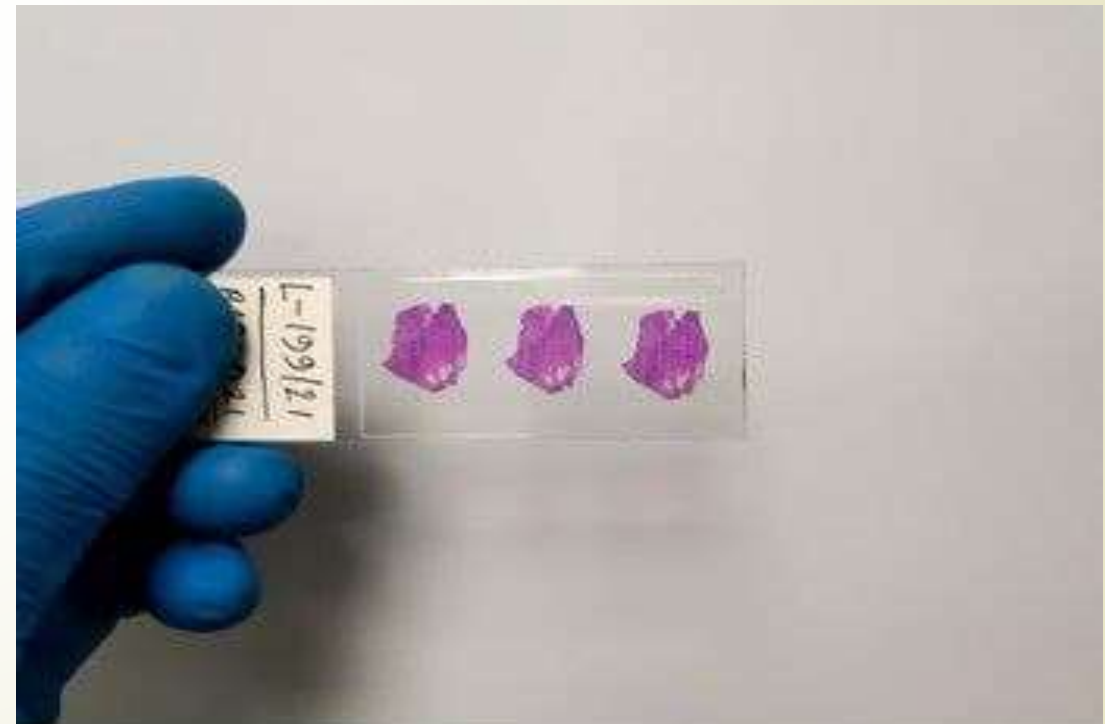


Microtome for tissue sectioning

H&E staining



H&E stain slides





Frozen section

processing involves:

- Freezing the tissue.
- Generating thin frozen slices of the specimen.
- slices of tissues are mounted onto glass slides.



Cryostat for frozen section





Cytopathology

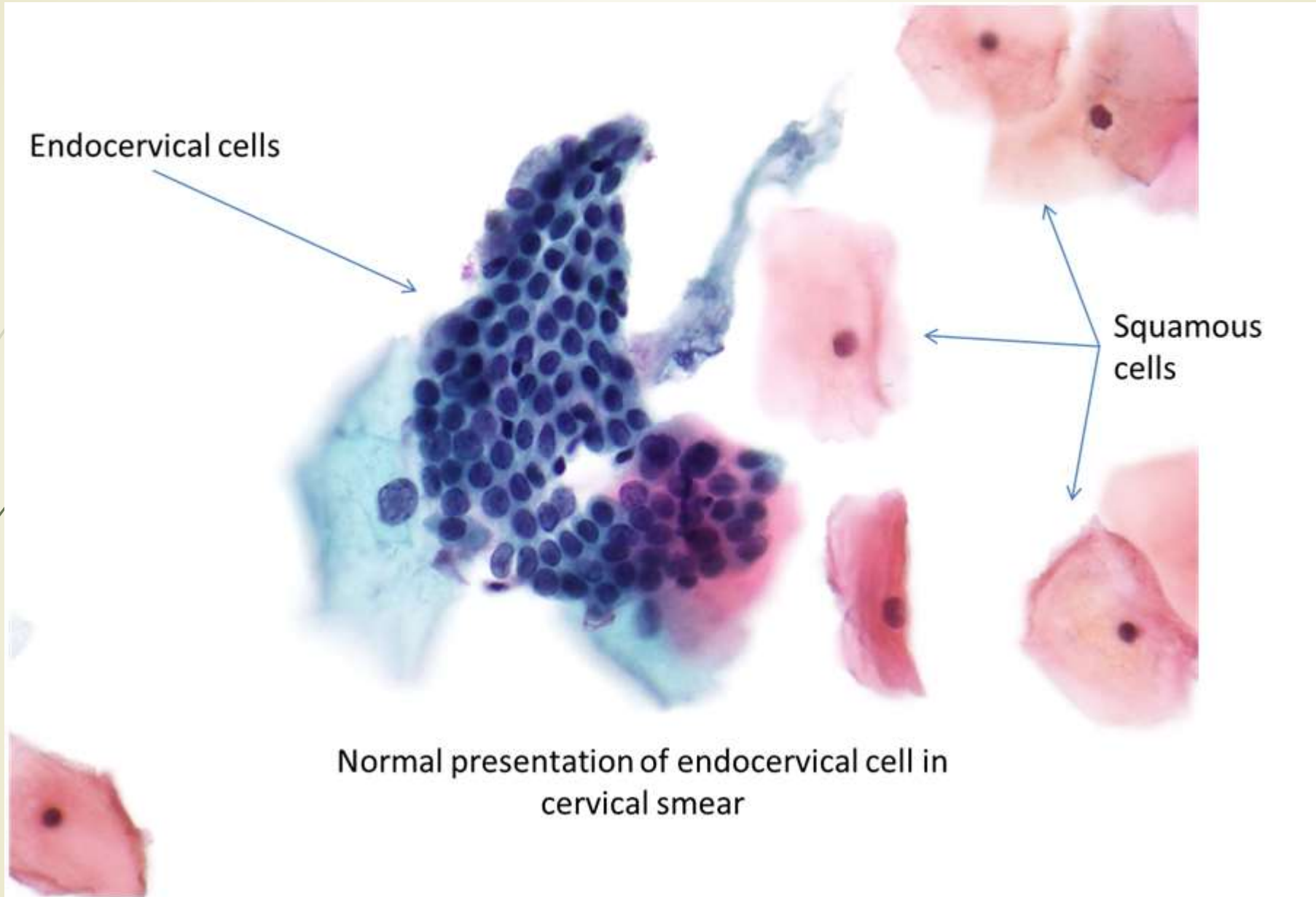
-Cytopathology is the use of specialist diagnostic techniques to examine individual cells extracted from tissues to determine the cause and nature of a disease.

- 1.Exfoliated cytopathology which deals with spontaneous cell shedding like cells from urine and sputum.
- 2.Fine needle aspiration in which the cells are extracted by the aid of negative pressure aspiration like thyroid gland and breast.

Endocervical cells

Squamous
cells

Normal presentation of endocervical cell in
cervical smear



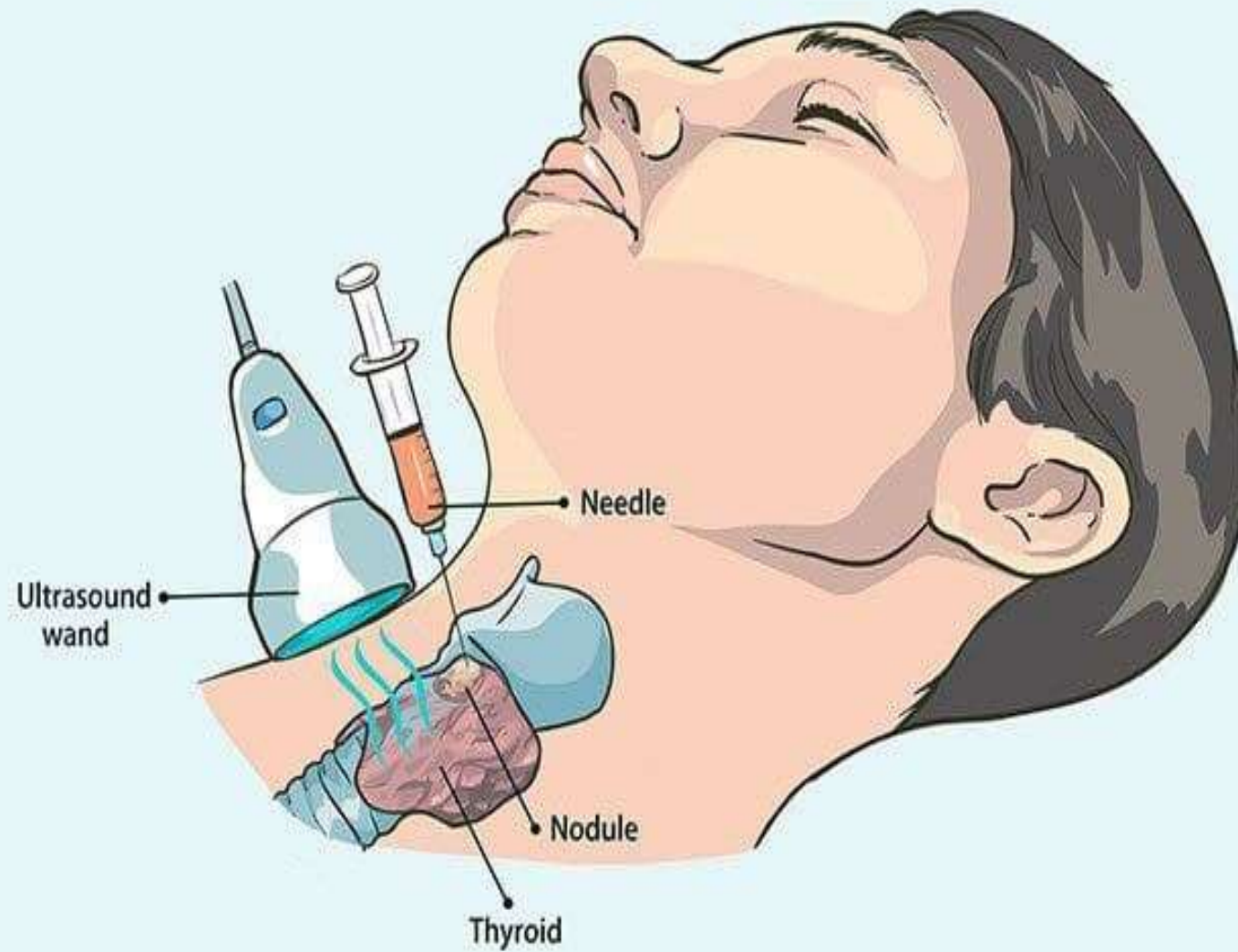


Fine needle aspiration cytology (FNAC)

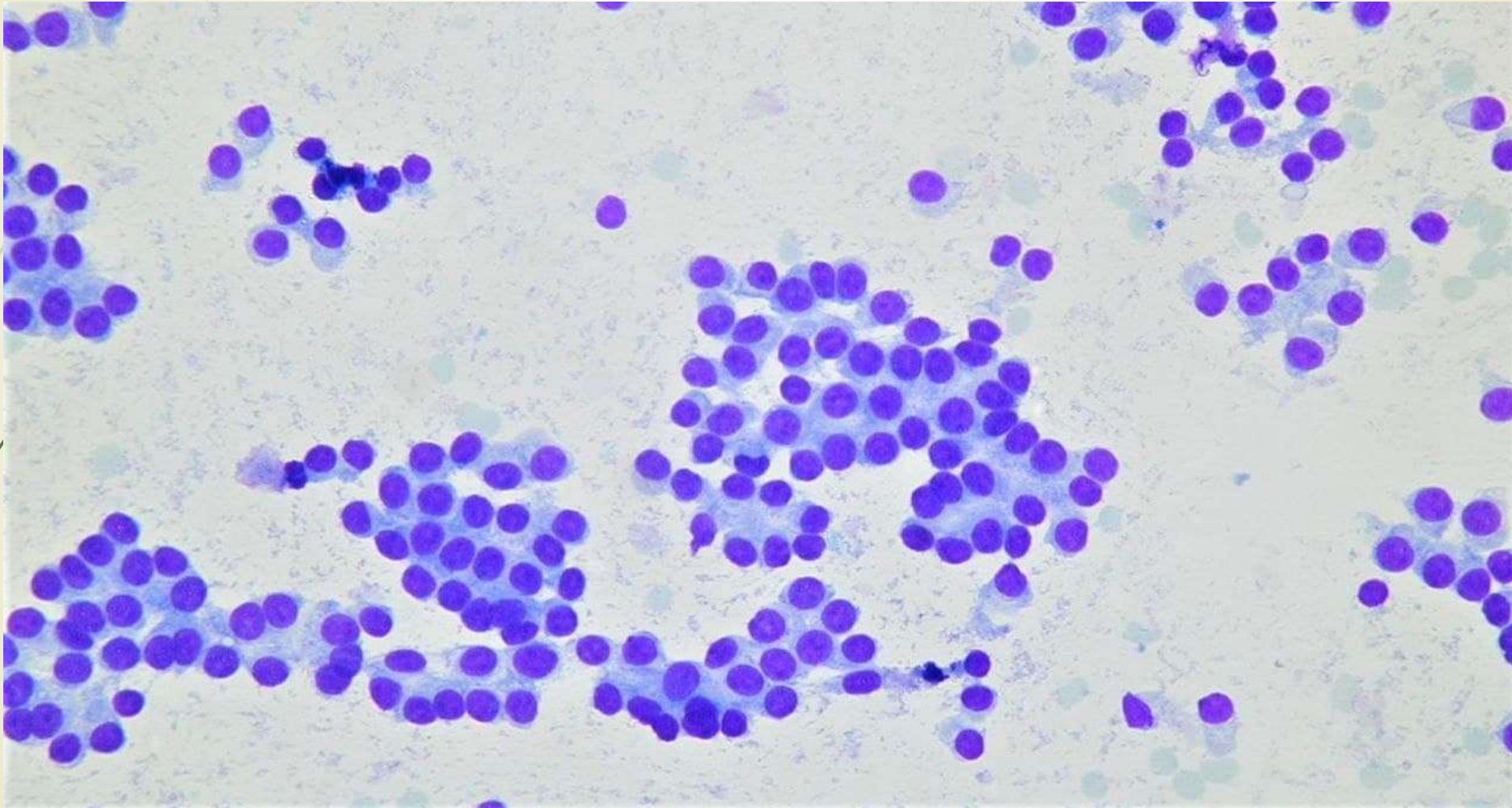
Use a narrow gauge (25-22G) needle to collect a sample of a lesion for microscopic examination.

- It allows a minimally invasive, rapid diagnosis of tissue
- Not preserve its histological architecture.

FNA technique



Cytology slide under microscope





THANK YOU